

The value added of machine learning to causal inference: evidence from revisited studies

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Summary: A new and rapidly growing econometric literature is making advances in the problem of using machine learning methods for causal inference questions. Yet, the empirical economics literature has not started to fully exploit the strengths of these modern methods. We revisit influential empirical studies with causal machine learning methods aiming to connect the econometric theory on these methods with empirical economics. We focus on the double machine learning, causal forest, and generic machine learning methods, in the context of both average and heterogeneous treatment effects. We illustrate the implementation of these methods in a variety of settings and highlight the relevance and value added relative to traditional methods used in the original studies.

Keywords: *Average treatment effects, causal inference, heterogeneous treatment effects, machine learning.*

JEL codes: C01, C21, D04.

1. INTRODUCTION

One of the key goals of empirical research in economics is to estimate the causal effect of a variable of interest on a targeted outcome. To avoid biases in the coefficients of interest due to omitted variables, particularly in observational studies, it is often desirable to include in the regressions a large number of controls. Even if the number of raw covariates is relatively small, including interactions and transformations can quickly increase the number of controls in the regression.

Machine learning (ML) methods can potentially be useful in such settings. However, *standard* ML prediction models are aimed at fundamentally different problems than most of the empirical work in economics. ML methods are designed and optimized for predicting the outcome in a test sample. Thus, a model is selected by optimizing the goodness of fit on the held-out test set. In contrast, in empirical economic research, the goodness of fit of a model is oftentimes reduced when estimating a causal effect, and the predictive accuracy is sacrificed in order to learn more deeply about a fundamental relationship that can guide policy decisions and counterfactual predictions (Athey and Imbens, 2019). These fundamental differences will eventually generate

biased estimates if *standard* ML techniques, designed for prediction, are used in the context of causal inference.¹ Nevertheless, a new and rapidly growing econometric literature is making advances in the problem of using ML methods for causal inference questions (see, e.g., Athey et al., 2018; Chernozhukov, Chetverikov et al., 2018; Chernozhukov, Demirer et al., 2018; Wager and Athey, 2018). This literature brings in new insights and theoretical results that are novel for both the ML and the econometrics/statistics literature. Despite these advances, the empirical economics literature has not started yet to fully exploit the strengths of these new modern causal inference methods.

The aim of this paper is to present empirical researchers evidence regarding the merits of causal machine learning methods in realistic settings. To this end, we revisit a number of influential papers by applying causal ML methods and compare the results with the traditional methods used in the original studies. In our analysis, we focus on both the average treatment effect (ATE) and heterogeneous treatment effects (HTE). Our main contribution is to illustrate how causal ML methods can be implemented in a variety of settings, and to highlight the relevance and additional gains that causal machine learning methods bring to the table relative to the standard econometric approaches. We further support some of our main findings with several Monte Carlo simulations, where the true data generating process is known. This allows us to compare the finite sample performance of causal ML estimators with traditional estimators in settings similar to the revisited studies.

When interested in the ATE, we employ the double/debiased machine learning (DML) method of Chernozhukov et al. (2017); when the focus is on heterogeneous treatment effects (HTE), we work with the causal forest method of Athey et al. (2019) and Wager and Athey (2018), and with the generic machine learning method for heterogeneous treatment effects developed by Chernozhukov, Demirer et al. (2018). These are newly developed causal machine learning methods with well-established theoretical properties. We re-examine a set of relatively recent influential studies that span a variety of topics in applied economics, published in the following journals: *The Quarterly Journal of Economics*, *American Economic Journal: Macroeconomics*, and *American Economic Journal: Applied Economics*. We choose papers for which the full replication data set is available either on the journal's website or on the authors' website. For the ATE, we revisit two observational studies: the study of Djankov et al. (2010a) on the effect of corporate taxes on investment and entrepreneurship, and the paper by Nunn and Treffer (2010a) on the effect of skill-biased tariffs on long-term economic growth. For the HTE, we select one observational study and one randomized control trial: we extend the observational study by DellaVigna and Kaplan (2007a), which investigates the effect of Fox News on the Republican vote share, and the analysis by Loyalka et al. (2019a) on the effect of a teacher training randomized intervention on student performance. All these papers include careful econometric analyses of the main research question and mechanisms, which we do not aim to re-examine in full. We instead focus on analysing the main questions.

Based on our results from the sample of revisited papers, we derive and systemize four main reasons why causal machine learning methods are relevant for causal analysis and add value relative to the traditional methods. These are general reasons that are not only valid for the specific settings or datasets of the papers that we revisit.

First, causal ML methods are powerful tools in using data to recover complex interactions among variables and flexibly estimate the relationship between the outcome, the treatment, and

¹ One of the underlying reasons is that, for instance, high dimensional regression adjustments such as lasso, ridge, elastic net, etc., shrink the estimated effects by construction, and ignoring this shrinkage will lead to biased treatment effect estimates.

the covariates. This feature is key when drawing inference based on the assumption that the treatment is unconfounded conditional on the observables, as in the case of most of the revisited studies, since this assumption is not testable. As some covariates can be correlated with both the treatment variable and the outcome, failing to condition on all relevant confounders may lead to biased estimates for the treatment effect. For example, for the effect of corporate taxes on investment and entrepreneurship, the original analysis in Djankov et al. (2010a) shows a negative and significant effect of corporate taxes on investment and entrepreneurship, but the authors show that these results do not survive when conditioning on all the potential controls at once. However, when implementing DML, we obtain larger estimates compared to Djankov et al. (2010a), which are often statistically significant. Furthermore, our analysis of the effect of skill-biased tariffs on growth suggests a smaller effect compared to Nunn and Trefler (2010a), which is often not statistically significant. We thus argue that the DML estimates are more robust to potential nonlinear confounders.²

Second, causal ML methods can be more suited than traditional methods when the number of covariates is large relative to the sample size, as they assume that the model is sparse (i.e., only a small number of covariates are relevant), and they use regularized regressions. For instance, in the study by Djankov et al. (2010a) and in some of the specifications in Nunn and Trefler (2010a), the number of raw covariates is large compared to the sample size, thus taking into account all possible nonlinear terms, such as interactions and transformations, would not be possible when using traditional methods. Indeed, no nonlinear terms other than logarithms are considered in Nunn and Trefler (2010a), and no nonlinear terms are included in Djankov et al. (2010a). In contrast, by using the DML method we ensure that our results take into account all potentially relevant confounders at once, both linearly and nonlinearly.

Third, the use of causal ML methods allows for *systematic model selection*. Many ML methods search for the best functional forms by estimating and comparing a wide range of alternative model specifications; the model selection is thus data-driven and fully documented. For example, our results for the effect of corporate taxes, originally explored by Djankov et al. (2010a), show that the data-driven model selection implemented by DML, which keeps a smaller set of influential confounding factors from among a large set of potential controls, leads to larger coefficients in absolute value and lower standard errors compared to ordinary least squares (OLS) regressions where all the covariates are included. With the traditional approach to model selection, uncertainty about the correct specification of the model can lead to choices that are relatively ad hoc; different specifications may lead to different point estimates, which in turn may lead to different policy decisions. Moreover, we further illustrate how these methods are also very useful tools for *supplementary analyses* or *robustness checks*. Typically, supplementary analysis is performed by presenting a number of selected regression specifications, while the approach of causal ML methods is more systematic, and ensures that important transformations of covariates that are not considered relevant a priori are not missed. For instance, we can consider our analysis of Nunn and Trefler (2010a) as a robustness check, as with DML we control for a data-driven function of the covariates. In this case, our results are different from the original analysis and statistical significance is lost.

Finally, causal machine learning methods prove to be very useful when one is interested in estimating heterogeneous treatment effects. As causal ML methods can handle many covariates

² It is important to note here that the idea of estimating treatment effects without making parametric assumptions about the way in which the covariates enter the equation has already been considered in the semi-parametric econometrics literature. See the review paper of Imbens and Wooldridge (2009) and Imbens and Rubin (2015). However, in practice, these semi-parametric kernel methods quickly break down if they have to deal with more than a few covariates.

potentially responsible for treatment effect heterogeneity in a systematic way, it is less likely that relevant heterogeneous effects will be missed, compared to manually modelling different interaction terms. This feature is exemplified by our analysis of the heterogeneous effects of Fox News on the Republican vote share first explored by DellaVigna and Kaplan (2007a) and of the teacher training intervention studied by Loyalka et al. (2019a): our results reveal drivers of heterogeneity that were unexplored in the original analysis. In addition, causal ML methods tailored for estimating heterogeneous treatment effects provide valid confidence intervals in high dimensional settings, as opposed to traditional methods where standard p -values for single hypothesis testing are not reliable.³ This is due to the multiple hypotheses testing problem, which can occur when researchers search iteratively for treatment effect heterogeneity, over a large number of covariates.^{4,5}

Our main findings from the revisited studies are supported by several Monte Carlo simulations inspired by empirically relevant settings. Our focus is on evaluating the relative finite sample performance of traditional and causal machine learning methods: (i) when the relationship between the outcome and the covariates as well as the treatment and the covariates is either linear or nonlinear; and (ii) when the number of covariates used in estimation increases relative to the sample size. To this end we focus on the DML and show that it outperforms OLS when the true nuisance relationship is nonlinear. Moreover, we find that the performance of DML relative to OLS improves as the number of covariates increases relative to the sample size, in both the linear case and the nonlinear case.

The econometric theory literature on adapting standard machine learning techniques to causal inference questions is by now fast growing. See, for example, Chernozhukov et al. (2017), Athey et al. (2018), Chernozhukov, Chetverikov et al. (2018), Colangelo and Lee (2020), and Farrell et al. (2021) for the ATE; and Athey and Imbens (2016), Chernozhukov, Demirer et al. (2018), Semenova et al. (2018), Wager and Athey (2018), Athey et al. (2019), and Oprescu et al. (2019) for the HTE. In the statistics literature, estimation of ATE and HTE with machine learning methods has been the focus in Zeileis et al. (2008), Su et al. (2009), Hill (2011), Van der Laan and Rose (2011), and Imai and Ratkovic (2013), among others. A few papers started employing the above-mentioned methods in interesting early applications. See, for example, Bertrand et al. (2017), Davis and Heller (2017), Strittmatter (2019), Davis and Heller (2020) and Knaus et al. (2022) for the causal forest, and Deryugina et al. (2019) for the generic machine learning.

In what follows, we present our main findings on average treatment effects using double machine learning in Section 2. The analysis of heterogeneous treatment effects using the causal forest and the generic machine learning method are described in Section 3. In Section 4 we summarize our main takeaways and recommendations for the applied researchers interested in employing these methods. An intuitive description of the methodology, more details on the

³ Note that the causal forest method by Wager and Athey (2018) is not developed for very high dimensional settings; however, the generic machine learning method of Chernozhukov, Demirer et al. (2018) can handle a large number of covariates.

⁴ While solutions have been proposed to correct for the issue of multiple hypothesis testing (for example, List et al., 2019), when the number of covariates is large, the power of these approaches to detect heterogeneity is low (Athey and Imbens, 2017).

⁵ A related issue is the *ex post* selection of significant heterogeneous effects. To avoid this problem, in randomized control trials researchers are often required to specify before the experiment which heterogeneous effects they are interested to look into, in order to avoid searching for, and only reporting, significant effects. However, this limits the ability of the researcher to find unexpected relevant heterogeneity. Causal ML methods ensure that relevant heterogeneity is not missed while also providing valid confidence intervals. In addition, in observational studies, where pre-analysis plans are not common practice, causal ML methods can be particularly useful.

revisited papers, details on the implementations of the methods, and the results of the Monte Carlo study are deferred to in the [Online Appendix](#).

2. AVERAGE TREATMENT EFFECTS

This section contains the analysis on the ATE for the effect of corporate taxes on investment and entrepreneurship (Djankov et al., 2010a) and the effect of skill-biased tariffs on growth (Nunn and Trefler, 2010a) using the double machine learning method (Chernozhukov et al., 2017).⁶

2.1. *The Effect of corporate taxes on investment and entrepreneurship*

2.1.1. Description of original analysis. The first paper that we revisit using causal machine learning methods investigates the relationship between corporate taxes on investment and entrepreneurship (Djankov et al., 2010a). This is an observational study that shows a negative effect of corporate taxes on investment and entrepreneurship, by estimating OLS country-level regressions with different measures of corporate tax rates for the year 2004. The sample includes a set of 50–85 countries, depending on the specification. In the original paper, four outcome variables are examined: investment as a percentage of GDP, FDI as a percentage of GDP, business density per 100 people, and the average entry rate. Three measures of corporate taxes are considered: statutory corporate tax rates, actual first-year corporate income tax liability of a new company, and the tax rate which takes into account actual depreciation schedules going five years forward.

The original paper reports the results for several regression specifications with different sets of control variables, to account for potential confounders that correlate with corporate tax rates, and are also determinants of the outcomes.⁷ Djankov et al. (2010a) present regression results where the first three sets of covariates are added separately. A final robustness check includes all control variables (twelve in total) in the same regression. In the specifications which include only one set of controls at a time, the paper shows a negative and statistically significant effect of corporate taxes on entrepreneurship and investment. However, when adding all the controls, the relationship is still negative, but the coefficients are smaller in size and no longer statistically significant.

2.1.2. DML analysis. We revisit the final robustness check of the paper, which includes all four sets of covariates at the same time, using the DML partially linear model. Table 1 presents the results. Columns (1) to (7) display the DML point estimates for the effect of corporate taxes on investment and entrepreneurship, using different ML methods to estimate the nuisance functions. Further details on how the DML estimates are obtained, the methods used, and the tuning parameters are described in Section S2.1 of the [Online Appendix](#).

We notice that all the DML point estimates have negative signs and generally similar magnitudes across the ML methods. Compared to the original paper results with the full set of covariates, reported in column (8), the magnitude of the DML coefficients is higher in absolute value, and the

⁶ For our analysis, we use the replication data provided by the authors Djankov et al. (2010b) and Nunn and Trefler (2010b).

⁷ The first set of controls includes measures of other taxes; the second set includes measures for the number of other tax payments made and for tax evasion; the third set includes measures for institutions; the fourth set includes measures of inflation. Section S2.1 of the [Online Appendix](#) includes more details on the regressions estimated in Djankov et al. (2010a) and describes the control variables.

Table 1. The effect of corporate taxes on investment and entrepreneurship.

	(1) Lasso	(2) Reg. Tree	(3) Boosting	(4) Forest	(5) Neural Net.	(6) Ensemble	(7) Best	(8) OLS
<i>Panel A: Investment 2003–2005</i>								
Statutory corporate tax rate	−0.081 (0.083)	−0.056 (0.075)	−0.065 (0.076)	−0.077 (0.084)	−0.056 (0.103)	−0.074 (0.09)	−0.068 (0.089)	−0.064 (0.098)
First-year effective tax rate	−0.122 (0.092)	−0.133 (0.089)	−0.156 (0.087)	−0.142 (0.093)	−0.137 (0.101)	−0.134 (0.091)	−0.138 (0.091)	−0.117 (0.106)
Five-year effective tax rate	−0.178 (0.096)	−0.179 (0.095)	−0.199 (0.091)	−0.204 (0.094)	−0.218 (0.101)	−0.195 (0.099)	−0.203 (0.101)	−0.189 (0.118)
Observations	61	61	61	61	61	61	61	61
<i>Panel B: FDI 2003–2005</i>								
Statutory corporate tax rate	−0.136 (0.085)	−0.167 (0.088)	−0.142 (0.09)	−0.131 (0.091)	−0.078 (0.09)	−0.123 (0.092)	−0.112 (0.092)	−0.030 (0.066)
First-year effective tax rate	−0.172 (0.091)	−0.203 (0.084)	−0.188 (0.085)	−0.169 (0.079)	−0.154 (0.084)	−0.168 (0.088)	−0.16 (0.085)	−0.1 (0.071)
Five-year effective tax rate	−0.162 (0.093)	−0.183 (0.076)	−0.169 (0.076)	−0.177 (0.08)	−0.164 (0.09)	−0.17 (0.086)	−0.15 (0.084)	−0.095 (0.081)
Observations	61	61	61	61	61	61	61	61
<i>Panel C: Business density</i>								
Statutory corporate tax rate	−0.054 (0.063)	−0.088 (0.072)	−0.063 (0.066)	−0.06 (0.063)	−0.031 (0.077)	−0.054 (0.067)	−0.042 (0.069)	−0.034 (0.083)
First-year effective tax rate	−0.105 (0.074)	−0.158 (0.087)	−0.123 (0.073)	−0.115 (0.07)	−0.091 (0.083)	−0.099 (0.074)	−0.102 (0.076)	−0.068 (0.092)
Five-year effective tax rate	−0.093 (0.075)	−0.14 (0.085)	−0.11 (0.072)	−0.104 (0.068)	−0.087 (0.086)	−0.107 (0.076)	−0.098 (0.075)	−0.070 (0.103)
Observations	60	60	60	60	60	60	60	60
<i>Panel D: Average entry rate 2000–2004</i>								
Statutory corporate tax rate	−0.128 (0.067)	−0.15 (0.066)	−0.141 (0.066)	−0.133 (0.065)	−0.079 (0.081)	−0.12 (0.071)	−0.113 (0.071)	−0.029 (0.086)
First-year effective tax rate	−0.107 (0.075)	−0.136 (0.066)	−0.14 (0.069)	−0.115 (0.066)	−0.109 (0.082)	−0.116 (0.074)	−0.112 (0.072)	−0.083 (0.094)
Five-year effective tax rate	−0.156 (0.076)	−0.146 (0.072)	−0.155 (0.072)	−0.15 (0.07)	−0.175 (0.087)	−0.155 (0.075)	−0.152 (0.077)	−0.133 (0.103)
Observations	50	50	50	50	50	50	50	50
Raw covariates	12	12	12	12	12	12	12	12

Notes: Analysis of Djankov et al. (2010a, tbl. 5D) using DML. Column 8 reports the original paper estimates. Standard errors are reported in parentheses. Standard errors adjusted for variability across splits using the median method are reported for the DML estimates. The number of covariates does not include the treatment variable.

standard errors are lower in most regressions. Additionally, the results are statistically significant, at least at the 10% level, in almost half (40 out of 84) of the regressions.

It seems that applying regularization here leads to lower standard errors and higher precision. However, in the absence of the known ground truth, whether the DML estimates are closer to the truth or not can be questioned. To offer further clarifications on this point, we note that both the original analysis and our analysis rely on the unconfoundedness assumption. In the case of the OLS analysis, the (implicit) assumption is that it is sufficient to control for all factors linearly. However, DML allows for a more flexible estimation, including potential nonlinear confounders as well as linear controls. This means that DML allows to relax the original assumption and replaces it with a weaker assumption, i.e., that the effect of confounders can be sufficiently controlled for by including the same controls as in the original analysis both linearly and nonlinearly. Furthermore, one might be interested in investigating what are these nonlinear terms that make the estimates different. However, this can be a challenging task when ML methods (such as neural networks, hybrid methods, etc.) are used to estimate the nuisance functions. What can potentially be done is

analysing the lasso coefficients that are not shrunk to zero and looking for nonlinearities among these. As an example, we show in [Figure S3.1](#) in the [Online Appendix](#), the most relevant among the nonlinear terms selected by the lasso, for one of the DML regressions reported in [Table 1](#).⁸ Here, we note that some nonlinear terms appear in both the treatment nuisance function $\hat{m}(\cdot)$ and in the outcome nuisance function $\hat{g}(\cdot)$.⁹ This is suggestive of the fact that there are nonlinearities that are correlated with both the treatment variable and the outcome. These were missed by the analysis in the original paper, and their omission could lead to biased coefficients of the corporate taxes variables. In this case, controlling for all relevant confounders strengthens the main results of the original analysis: in many cases the DML treatment effect estimates are larger in absolute value, and statistically significant. The results of our Monte Carlo simulations, presented in [Section S4](#) of the [Online Appendix](#), further highlight the relevance of using DML in the presence of nonlinearities, even in small sample sizes.

The DML results are obtained by tuning the parameters of the ML methods via cross-validation, whenever this is theoretically justifiable. Some of the parameters, however, are not data-driven (for example, the number of trees or the leaf node size). Thus, we perform additional sensitivity checks on the values used for these non-adaptive tuning parameters. In addition, we change the activation function and vary the number of layers in the neural net. The results, not reported, but available on request, are consistent with those reported here.

The good performance of causal machine learning methods is subject to the assumption of *sparsity*. However, the sparsity assumption is not testable and thus it must be used with caution. In our empirical applications, it is reassuring, however, that the results obtained from the different ML methods give very close second-stage DML estimates (the estimates of the ATEs). This is consistent with the existence of a sparse basis which is concomitantly captured by all ML methods.

This empirical application is a good example to illustrate the usefulness of causal ML methods in the *typical trade-off* that applied researchers often face. On the one hand, the researcher wants to control for as many potential confounders as possible, in order to improve the credibility of the unconfoundedness assumption. On the other hand, naively controlling for a large set of covariates, especially when the sample size is small, can lead to imprecise estimates and larger standard errors. Notice that in this example, the authors implement a ‘kitchen sink’ regression and control for all the covariates at once, resulting in larger standard errors than the ones that we obtain. The DML method helps with this trade-off by improving the credibility of the unconfoundedness assumption (as it captures the effect of confounders more flexibly), but, at the same time, it implements a data-driven variable selection technique to keep a smaller set of influential confounding factors from among a large set of potential controls, thus resulting in lower standard errors.

Lastly, to further support our obtained causal ML results, we also perform the analysis and compute the ATEs using the causal forest.¹⁰ We report the causal forest estimates in [Table S3.2](#) in the [Online Appendix](#). The results are consistent with the DML estimates.¹¹

⁸ It is important to note here that we do not make inference using the lasso coefficients, but we analyse the magnitude of the coefficients as a measure of the covariates’ importance for predicting the outcome and the treatment variables.

⁹ Further details about the lasso coefficients analysis are reported in [Section S2.1](#) of the [Online Appendix](#).

¹⁰ See [Section S1.2](#) in the [Online Appendix](#) for a description of the causal forest method. We consider the causal forest, and not the generic method developed by Chernozhukov, Demirer et al. (2018), as the latter requires a binary treatment variable.

¹¹ [Table S3.2](#) in the [Online Appendix](#) shows the results using default values of the parameters (which are reported in the notes of the table). Due to the small sample size, we are unable to tune the parameters with cross-validation; thus, we perform sensitivity analysis varying the parameter values. The results, available on request, are consistent with those reported in [Table S3.2](#) in the [Online Appendix](#).

2.2. The effect of skill-biased tariffs on growth

2.2.1. Description of original analysis. The study by Nunn and Trefler (2010a) investigates the relationship between skill-biased tariffs, i.e., a tariff structure that disproportionately favours skill-intensive industries, and long-term economic growth. The authors develop a theoretical framework based on Grossman and Helpman (1991) that shows how tariffs that focus on skill-intensive industries can lead to a disproportionate expansion of skill-intensive industries, which then leads to higher long-term growth. Furthermore, using both cross-country and industry-level data, the paper provides evidence of a positive relationship between the two variables, and delves into the mechanisms of this relationship. The findings suggest that the mechanisms from the theoretical framework can explain only part of the total correlation between skill-biased tariffs and growth. The paper attributes the remaining part of the correlation to the endogeneity of skill-biased tariffs, and in particular to the relationship between institutions and the skill-bias of tariffs: countries with good institutions tend to protect more skill-intensive industries.

In Nunn and Trefler (2010a), three measures of the skill-bias of tariffs in the initial time period are used:¹² the correlation between the industry tariffs and the industry's skill-intensity, and two measures based on the difference between the log average tariffs in skill-intensive industries and log average tariffs in unskilled-intensive industries, which use different cut-off values for industry skill-intensity. In the country-level estimates, the outcome is log annual per capita GDP growth, and the regressions include a set of control variables.¹³ The country-level regressions includes 63 observations.

For the industry-level estimates, the outcome variable is the average annual log change in industry output in each country, and the regressions include all the controls that appear in the country-level regressions, plus industry fixed effects. These regressions include 1,004 data points for 59 countries. An additional variable (the initial industry tariff) is included in some specifications to capture a potential mechanism: skill-biased tariffs can shift resources towards skill-intensive industries that generate positive externalities, thus leading to higher long-term growth. Thus, industries that have higher initial tariffs should have higher long-run output. If this channel can explain the effect of skill-bias on growth, the coefficient of the skill-bias of tariffs would decrease in size when this variable is included in the regression.

2.2.2. DML analysis. We revisit the country and industry-level regressions reported in Nunn and Trefler (2010a, tbl. 4 [cols. 1, 2, and 4], tbl. 5 [cols. 1, 2, and 4], tbl. 6 [cols. 1, 3, and 7]). Further details on how the DML estimates are obtained and on the tuning parameter values are reported in Section S2.2 of the Online Appendix.

Table 2 shows the results of the DML partially linear model using country-level data. The DML treatment effect estimates are considerably smaller than the original paper's across all ML methods and across the three different treatment variables. Moreover, the estimated effects are not statistically significant, except the coefficients estimated using the lasso (estimates using boosting in panel B and ensemble in panel C are also significant at the 10% level). Additionally, we report the DML results using the industry-level data set (Table S3.3 and Table S3.4 in the Online Appendix show the results with and without including the initial industry tariff, respectively).

¹² The initial time period is 1972 for 21 countries, 1980–1983 for 30 countries and 1985–1987 for twelve countries. The end period is 2000 for most countries, except for three of them, for which data ends in 1996. See Nunn and Trefler (2010a, tbl. 1) for a list of the countries included and the respective time periods.

¹³ Further details on the regressions estimated by Nunn and Trefler (2010a) and on the control variables are described in Section S2.2 of the Online Appendix.

Table 2. The structure of tariffs and long-term growth: country-level estimates.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Lasso	Reg. Tree	Boosting	Forest	Neural Net	Ensemble	Best	OLS
<i>Panel A: Skill tariff correlation</i>								
Skill tariff correlation	0.019 (0.010)	0.016 (0.012)	0.016 (0.011)	0.016 (0.011)	0.013 (0.015)	0.019 (0.012)	0.016 (0.011)	0.035 (0.010)
<i>Panel B: Tariff differential (low cut-off)</i>								
Tariff differential (low cut-off)	0.010 (0.005)	0.008 (0.005)	0.009 (0.005)	0.008 (0.006)	0.006 (0.008)	0.008 (0.006)	0.008 (0.006)	0.016 (0.006)
<i>Panel C: Tariff differential (high cut-off)</i>								
Tariff differential (high cut-off)	0.009 (0.005)	0.006 (0.005)	0.007 (0.005)	0.008 (0.005)	0.013 (0.008)	0.009 (0.005)	0.008 (0.005)	0.02 (0.004)
Observations	63	63	63	63	63	63	63	63
Raw covariates	17	17	17	17	17	17	17	17

Notes: Analysis of Nunn and Trefler (2010a, tbl. 4, cols. 1, 2, 4) using DML. Column (8) reports the original paper estimates. Standard errors are reported in parentheses. Standard errors adjusted for variability across splits using the median method are reported for the DML estimates. The number of covariates does not include the treatment variable.

Similarly to the country-level estimates, the industry-level estimates are not statistically significant across all methods, except for the boosting estimates.

Overall, the DML results suggest that the correlation between skill-biased tariffs and long-term economic growth is not robust to controlling for an unknown function of the average tariff level, country characteristics, initial production structure, and cohort and region fixed effects. Indeed, the fact that the DML estimates are insignificant points to the presence of nonlinear confounding effects that are not accurately captured by the OLS regressions.

It is worth noting here that the original paper attributes most of the correlation found between the treatment variables and long-term growth to the endogeneity of the skill-biased tariff variables, arising from the fact that skill-biased tariffs are more likely in countries with better institutions. Interestingly, in this example the country-level DML estimates are in line with the notion that the direct effect of the skill-bias of tariffs is smaller than what is estimated by the OLS regressions. Finally, our results only concern the relationship between skill-biased tariffs and long-run economic growth, and not the relationship between skill-biased tariffs and institutions, or between institutions and long-run growth, which are examined in the original paper. Thus, our findings are consistent with the alternative mechanism described in Nunn and Trefler (2010a), i.e., the existence of a causal relationship between institutions and economic growth.

Furthermore, we estimate the country-level regressions with the causal forest method. The ATEs obtained with the causal forest, reported in Table S3.5 in the Online Appendix, are very similar to the DML estimates.¹⁴

3. HETEROGENEOUS TREATMENT EFFECTS

This section focuses on the analysis of HTE for the effect of Fox News on Republican voting (DellaVigna and Kaplan, 2007a) using the causal forest method (Wager and Athey, 2018; Athey

¹⁴ As in the first application, the values of the tuning parameters used are the default values, and they are reported in the notes of Table S3.5 in the Online Appendix. Results considering different values for the parameters are consistent with those reported and are available on request.

et al., 2019) and the effect of a teacher training intervention (Loyalka et al., 2019a) using the generic machine learning method (Chernozhukov, Demirer et al., 2018).¹⁵

3.1. The effect of Fox News on the republican vote share

3.1.1. Description of original analysis. In this section we revisit and further analyse the study by DellaVigna and Kaplan (2007a). This paper examines the impact of media bias on voting outcomes. Specifically, it analyses the impact of the entry of a conservative cable television channel, Fox News, on the Republican Party's vote share in the United States. To identify the causal effect of Fox News on voting, the authors investigate whether towns where Fox News became available between 1996 and 2000 experienced an increase in the vote share for the Republican Party in presidential elections during the same time period. The estimation is performed on a data set at the town level, comprising information on 9,256 towns.

We consider the main outcome variable, i.e., the change in the vote share for the Republican Party between 1996 and 2000. The treatment variable is a dummy indicating whether Fox News had become available between 1996 and 2000. To capture potential confounders, a number of control variables are included in the regressions.¹⁶

DellaVigna and Kaplan (2007a) find a positive effect of Fox News on the Republican vote share. Moreover, they explore heterogeneity along a selected set of town characteristics: the number of available cable channels, the share of urban population, and whether the town is in a swing or Republican district. They do this by adding to the regression interaction effects of these covariates with the treatment variable.¹⁷

3.1.2. Causal forest analysis. We perform the HTE analysis using the causal forest method. Exploring heterogeneous effects is important for this study, in order to understand whether there are town or district characteristics that act as effect modifiers. While the average effects are informative for the impact of Fox News on the whole sample, it is often the case that treatment effects are not homogeneous. It is possible that the effect of Fox News was concentrated in some areas only. Understanding better the characteristics of the areas which saw the strongest and weakest responses can shed light on the mechanisms. The aim of this exercise is two-fold. First, we take an agnostic view about the nature of heterogeneity, and we investigate whether there are town or district characteristics which are treatment effect modifiers. Second, we examine whether the HTE analysis from the original paper matches the results from the causal ML methods.

We focus on one of the two preferred specifications from the original paper: the one that includes district fixed effects. We present results for two versions of the causal forest, which account for district-level effects in different ways. In the first set of results, we include in the analysis dummy variables indicating the congressional district where the town is located. In the second set of results, we implement a cluster-robust version of the random forest developed by Athey and Wager (2019), where we treat each district as a separate cluster. The advantage of the cluster-robust causal forest is that it does not assume that clusters have an additive effect on the outcome. Further details on the clustered-robust causal forest and tuning parameter values used for the analysis are discussed in Section S2.3 of the Online Appendix. Note that pointwise

¹⁵ For our analysis, we use the replication data provided by the authors DellaVigna and Kaplan (2007b) and Loyalka et al. (2019b).

¹⁶ Further details on the regressions and on the control variables in DellaVigna and Kaplan (2007a) are described in Section S2.3 of the Online Appendix.

¹⁷ The findings are reported in DellaVigna and Kaplan (2007a, tbl. 6 of the original paper).

Table 3. Fox News—Causal forest: average treatment effects and test for heterogeneity.

	(1) District dummies	(2) Cluster-robust
Fox News effect (ATE)	0.0065 (0.0016)	0.0065 (.0026)
Fox News effect above median	0.011 (0.0023)	0.0078 (0.0028)
Fox News effect below median	− 0.0028 (0.0022)	0.0034 (0.0042)
95% Confidence interval for the difference	(0.00759, 0.01985)	(− 0.00545, 0.01437)
Observations	9,256	9,256

Notes: This table reports the estimated average treatment effect and a test for overall heterogeneity using the causal forest. Standard errors are reported in parentheses.

asymptotic normality for the causal forest is provided for cases where the number of covariates is relatively low, and the covariates are continuous. To circumvent this issue, we perform a robustness check using the approach implemented by Athey and Wager (2019), where we train a preliminary random forest on all covariates, after which we run a final random forest on a reduced number of features. The results are discussed in Section S2.3 of the Online Appendix and are very similar to those presented in this section.

We begin by discussing the average treatment effect. The results are presented in Table 3. As in the original analysis, we find a positive and significant effect of Fox News on the Republican vote share, both when including district dummies and when implementing the clustered-robust causal forest; however, the standard error in the clustered forest is larger. Our results suggest that in towns where Fox News became available the Republican Party obtained a higher vote share by 0.65 percentage points on average, compared to towns where Fox News was not available. The ATE estimates are similar to the original paper estimates, which range between 0.4 and 0.7 percentage points (reported in DellaVigna and Kaplan, 2007a, tbl. 4, cols. 4–7).

Next, we want to assess whether the causal forest can recover heterogeneity of treatment effects. As pointed out in Athey and Wager (2019), we can group observations according to whether their estimated out-of-bag conditional average treatment effect (CATE) is above or below the median CATE, and we can estimate the average treatment effect separately for these two subgroups. These are reported in Table 3 as *Fox News effect above median* and *Fox News effect below median*. Note that these results should be interpreted with caution, as developing uniformly valid standard errors for the causal forest is still an open question and we do not adjust our inference for the fact that we use predictions that contain error to build our subgroups.¹⁸ The difference between the two subgroup estimates is large when including district dummies, suggesting that there is potential for heterogeneity, and it is statistically significant, as indicated by the fact that the 95% confidence interval for the difference between the two estimates does not contain zero (see column 1 of Table 3). However, the same heuristic test for the clustered-robust forest does not detect significant heterogeneity in the treatment effect. This could indicate that heterogeneity in the model with district dummy variables is overstated, because the dummy variables cannot

¹⁸ To supplement our analysis, we implement an additional test for overall heterogeneity, inspired by the best linear predictor method in Chernozhukov, Demirer et al. (2018). The results, reported in Table S3.6 and discussed in Section S2.3 of the Online Appendix, are in line with those obtained from the test in Table 3.

Table 4. Fox News—causal forest: HTE analysis.

	(1) CATE below median	(2) CATE above median	(3) <i>p</i> -value difference
<i>Panel A: District dummies</i>			
Employment rate, diff. btw. 2000 and 1990	0.00929 (0.00243)	0.0005 (0.00204)	0.00562
Share high school degree 2000	0.00806 (0.00222)	− 0.00032 (0.00216)	0.00676
Decile 10 in no. cable channels available	0.00872 (0.00191)	− 0.00456 (0.00262)	4e-05
<i>Panel B: Cluster-robust</i>			
Employment rate, diff. btw. 2000 and 1990	0.00939 (0.00258)	0.00013 (0.00412)	0.05676
Share high school degree 2000	0.0085 (0.00301)	− 0.0015 (0.00425)	0.05492
Decile 10 in no. cable channels available	0.0086 (0.00284)	− 0.00524 (0.00513)	0.01823

Notes: This table reports the effect of Fox News on the Republican vote share for towns with values below (column 1) and above (column 2) the median of each variable. Column 3 presents the *p*-value for the null of no difference between the estimates in columns 1 and 2. Standard errors are reported in parentheses.

appropriately capture the district-specific effects. The cluster-robust causal forest offers a more flexible way to capture district-specific effects, and may be more suitable in this case.¹⁹

Although the results of the test for overall heterogeneity are mixed, it is still possible for heterogeneity to be present along some of the covariates. Hence, we investigate whether any of the included covariates are possible sources of heterogeneity. To do this, for each variable, we split the sample in two parts, based on whether the value of the covariate of interest is below and above the median, and we estimate the average treatment effect for the two subsamples. Table 4 reports the HTE results along the variables that appear to be significant determinants of heterogeneity in both specifications, while Tables S3.7 and S3.8 in the Online Appendix report the results for the remaining variables. In addition, to gain further insight into which variables are more important for heterogeneity, we compute a measure of variable importance (see Athey and Wager, 2019).²⁰ Tables S3.9 and S3.10 in the Online Appendix report the variable importance measure for the covariates included in the district dummy variable specification and for the clustered-robust forest, respectively. We note that for both specifications, the variable importance measure is decreasing smoothly and we do not observe any variable that clearly stands out in terms of importance.

Our results in Table 4 show that three variables appear to be significant determinants of heterogeneity (at least at the 10% level) in both specifications: the change in employment between 1990 and 2000, the share of the population with education level equal to high school degree, and the 10th decile in number of cable channels available. We observe that the effect of Fox News on Republican voting is stronger in towns that experienced a smaller increase in the employment rate between 1990 and 2000. This finding may relate to the phenomenon of economic voting, i.e., the fact that voters tend to reward incumbents during periods of economic prosperity (e.g., Kramer, 1971; Fair, 1978; Pissarides, 1980; Lewis-Beck and Stegmaier, 2000). Areas that experienced lower economic growth (and a smaller increase in employment) may have been more easily persuaded to vote Republican in 2000, since prior to the presidential election of 2000 a Democratic

¹⁹ Athey and Wager (2019) find a similar result in their application, when comparing the causal forest without clustering with the cluster-robust version.

²⁰ See Section S2.3 of the Online Appendix for details on how this measure is constructed.

president (Bill Clinton) had been in power for two consecutive mandates. Moreover, we observe a larger effect of Fox News in towns where the share of population with education level equal to high school degree is below median. We also find a larger positive effect of Fox News in towns where the 10th decile in the number of cable channels is below median, while the effect is negative and insignificant in towns where this variable is above median.²¹

Next, we investigate whether the findings regarding heterogeneity from the original paper are confirmed with the causal forest. DellaVigna and Kaplan (2007a) found a larger effect of Fox News on the Republican vote share in towns with a smaller number of cable channels available when including district fixed effects. While we do not observe significant heterogeneity along this variable, our results for the 10th decile in the number of cable channels are in line with the findings of the original analysis, and, hence, suggest that the effect of Fox News diminishes in the presence of higher competition in cable channels. It is also interesting to note that the number of cable channels emerges as the variable with the highest importance score in both specifications, which further points to the importance of this variable for heterogeneity. When investigating heterogeneity along the political orientation of the district, we confirm the findings of DellaVigna and Kaplan (2007a): we observe no significantly different effect for swing districts, and we obtained mixed results for Republican districts, as we find a significantly smaller effect of Fox News in Republican districts (at the 10% level) when including district dummies, but not with the cluster-robust forest.²² However, in contrast to the original analysis, we do not find a significant difference in the effect of Fox News in rural versus urban towns, despite this being the only heterogeneity result that is robust in all specifications in DellaVigna and Kaplan (2007a).

In conclusion, our analysis of the HTE of Fox News on Republican voting confirms some of the findings from DellaVigna and Kaplan (2007a), namely the presence of heterogeneity along the number of cable channels and no robust heterogeneous effects for districts with different political orientations, but as opposed to the original paper it does not show different effects for urban and rural areas. The analysis with the causal forest further uncovers additional heterogeneity that was previously unexplored, such as a larger effect in towns that experienced a smaller increase in the employment rate, and a larger effect in towns with a lower share of population with high school degree. Finally, including district dummy variables results in the causal forest detecting more heterogeneity in treatment effects compared to the cluster-robust version, both when implementing the overall heterogeneity test and when analysing the HTE in terms of individual covariates. However, the model with district dummy variables could overstate the heterogeneity compared to the cluster-robust forest if the district dummies do not appropriately capture the district-specific effects. This points to the need of a more careful treatment of the issue of clustered observations when employing causal forests for empirical applications (Athey and Wager, 2019).

3.2. The effect of teacher training on student performance

3.2.1. Description of original analysis. We reanalyse a large-scale randomized experiment that investigates the effect of a teacher professional development (PD) programme in China on student achievement and on other student and teacher outcomes. The experiment was first studied by Loyalka et al. (2019a). Three hundred mathematics teachers, each employed in different schools across one province, took part in the intervention. The teachers were randomly assigned

²¹ The median value for the 10th decile in number of cable channels is zero; hence, towns with value of this variable above median correspond to towns that are in the top decile in terms of number of cable channels available.

²² DellaVigna and Kaplan (2007a) found mixed results for Republican districts in different specifications.

to one of the different treatment arms: PD only; PD plus a continuous follow-up with additional material and tasks for the trainees; PD plus an evaluation of the extent to which the teachers remembered the content of the training sessions; or no PD (control group). The PD intervention consisted of lectures and discussions.

Randomization was implemented at the school level, and in each school one teacher was nominated to participate in the intervention. The main results are obtained by estimating a cross-sectional regression, where the treatment variable is a dummy indicating the treatment arm that the school was assigned to. The data was collected at three points in time: at baseline, midline, and endline. Outcomes are measured at midline, or endline, and the main outcome of interest is student math achievement.²³ The control variables include student characteristics, teacher characteristics, and class size.²⁴

The original paper finds no significant effect of the PD intervention on students' achievement after one academic year, neither for the PD intervention alone, nor for the PD combined with the follow-up and/or the evaluation treatments. The authors also do not find any effect on other outcomes, such as teacher knowledge or student motivation. The lack of effectiveness of the programme is attributed to several factors: the content was too theoretical, the PD was delivered passively, and teachers could face constraints in the implementation of the suggested practices in the schools. Furthermore, the paper analyses heterogeneous treatment effects, by interacting the treatment variable with a number of student and teacher characteristics: student's household wealth, baseline achievement level, the amount of training the teacher has received prior to the intervention, student and teacher gender, whether the teacher has a college degree, and whether the teacher majored in math. The findings suggest that the effect of the treatment on students' achievement can differ by teacher characteristic; however, no heterogeneous effects are found in terms of characteristics of students.

3.2.2. Generic ML analysis. We extend the analysis of HTE conducted in the original paper, by implementing the generic machine learning method developed by Chernozhukov, Demirer et al. (2018). Exploring heterogeneous treatment effects is particularly relevant for this intervention, because a small and insignificant estimate for the ATE could hide significant heterogeneity. Our aim is to dig deeper into the analysis of heterogeneous treatment effects. First, we investigate whether there is significant heterogeneity in treatment effects; second, we analyse whether causal machine learning methods, by implementing a systematic search for heterogeneity across a large number of covariates, can offer additional insights about the characteristics of those who benefited from the programme and those who did not, compared to the traditional methods used in the original paper.

In our analysis, we focus on the main outcome of interest, i.e., student math achievement. Since the results in the original paper are consistently close to zero when comparing the three different treatment arms with the control group, we choose to only analyse one of the treatment arms, corresponding to the PD intervention plus the evaluation. The sample that we use includes 10,006 students in 201 schools. We follow Loyalka et al. (2019a) and cluster standard errors at the school level. In addition to the full set of controls included in the original paper, we also add to our analysis other variables that could be treatment effect modifiers: the baseline values of a

²³ As Loyalka et al. (2019a) show similar results when estimating the impact of the intervention at midline or endline, we focus on the outcome variables measured at endline.

²⁴ Section S2.4 of the [Online Appendix](#) describes the regressions and the control variables.

Table 5. Teacher training—generic method: best linear predictor.

	(1) ATE (β_1)	(2) HET (β_2)
Estimate	0.002	0.651
90% Confidence interval	(−0.068, 0.072)	(0.312, 0.990)
<i>p</i> -value	1.000	0.0003
Observations	10,006	10,006

Notes: The estimates are obtained using neural network to produce the proxy predictor $S(Z)$. The values reported correspond to the medians over 100 splits.

number of student-level variables, plus variables indicating teachers behaviour in the classroom, evaluated by students at baseline.²⁵

The generic method can be used in conjunction with a range of ML tools and Chernozhukov, Demirer et al. (2018) provide two measures—Best BLP and Best GATES to compare the performance of the different ML methods used for the estimation of the proxy predictors. We consider the following methods: elastic net, neural network, and random forest. Based on the results of the Best BLP and Best GATES analysis, reported in Table S3.13 of the Online Appendix, we choose to further work with the neural network.²⁶

We first analyse whether overall heterogeneity in treatment effects can be detected. We present results for the best linear predictor (BLP) of the CATE in Table 5. In line with the original paper, the estimated ATE, given by the coefficient β_1 , is small (the estimated impact of the PD is 0.002 SD) and not significantly different from zero. The estimated β_2 is instead large and significantly different from zero, which indicates that there is heterogeneity in treatment effects. Next, we estimate the group average treatment effects (GATES). We split the sample into five groups, based on the quintiles of the ML proxy predictor $S(Z)$. This analysis reveals further insights into the extent of heterogeneity. Table S3.14 in the Online Appendix reports the GATE in the top and bottom quintile and shows that the GATE in the top quintile is positive, whereas for the bottom quintile the estimated GATE is negative. Both estimates are statistically significant at the 10% level. The difference between the GATE for the top and the bottom quintile is significant, which confirms the presence of heterogeneity in treatment effects. Additionally, Figure 1 reports the GATES estimate and the 90% confidence interval for the five quintiles, as well as for the whole sample (the ATE is represented as a dark dashed line, and the confidence interval as two lighter dashed lines). Notice that for the three middle quintiles the effect of the teacher training intervention is not significantly different from zero.

We then turn to analysing the possible sources of heterogeneity, by implementing the classification analysis (CLAN). Thus, we analyse further the top and bottom quintile in terms of ATE, for which the effect of the PD intervention is positive and negative, respectively. In particular, we compare the student and teacher characteristics in the two groups. As a large number of covariates is available, we focus on the ten covariates for which the correlation with the proxy predictor, $S(Z)$, is highest, reported in Table 6. Table S3.15 in the Online Appendix shows the

²⁵ These additional variables are described in Section S2.4 of the Online Appendix. In Loyalka et al. (2019a), the baseline value of the outcome variable is included as a control. Hence, the baseline characteristics described above are not included in all regressions in the original analysis. However, we consider these characteristics as potential drivers of heterogeneity; therefore, we include the baseline values of all available variables in our heterogeneity analysis.

²⁶ Further details on the Best BLP and Best GATES measures and on the tuning parameters used in this analysis are discussed in Section S2.4 of the Online Appendix.

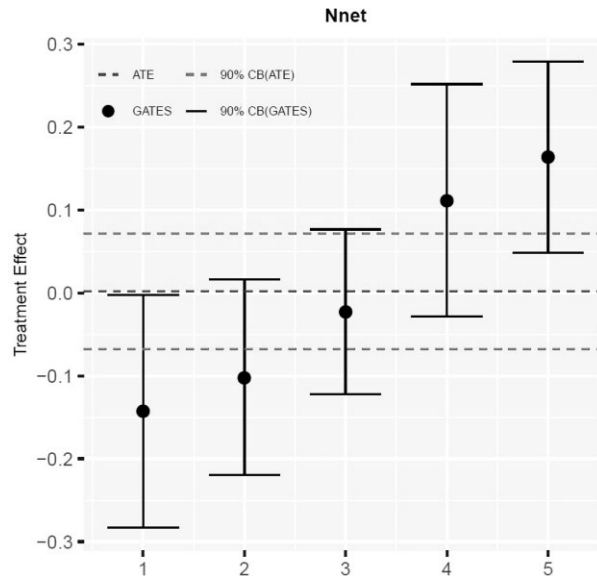


Figure 1. Teacher training—generic method: GATES.

Note: The estimates are obtained using neural network to produce the proxy predictor $S(Z)$. The point estimates and 90% confidence intervals correspond to the medians over 100 splits.

CLAN analysis for the remaining covariates. Table S3.16 in the Online Appendix reports the correlation for each of the covariates with $S(Z)$.

We start by analysing the *characteristics of the teachers* whose students belong to the least and most affected groups. Interestingly, the variable indicating whether the teacher has a college degree or not is the variable that is most correlated with the proxy predictor, and it was the only one among the variables tested which was found to be a treatment effect modifier across all treatment arms in the original paper. The students in the top quintile are more likely to be taught by a teacher who does not have a college degree, compared to the students in the bottom quintile. This is consistent with the results from Loyalka et al. (2019a), who found that the intervention has a negative effect on students whose teachers have a college degree, but a positive effect on students whose teachers are less qualified. Hence, the PD may help teachers who are less qualified, but, for more qualified teachers, the benefits of the intervention on their students do not outweigh the negative effect of the teachers being absent from the classroom in order to participate in the intervention. Whether or not the teacher majored in math is found to be a potential driver of heterogeneity with the generic method (the results are reported in Table S3.15 in the Online Appendix), whereas in the original paper it was not found to be significant when considering the effect of the PD plus evaluation, which we focus on.²⁷ The direction of the effect is consistent with what was found in the original analysis: the students in the top quintile are more likely to have been taught by a teacher who does not have a major in math, compared to the students in the bottom quintile. It is also interesting to note that the number of hours of

²⁷ When considering the PD plus follow-up, the authors find a significant negative effect on the scores of students whose teachers majored in math relative to the scores of those whose teachers did not.

Table 6. Teacher training—generic method: classification analysis.

	(1) 20% most affected	(2) 20% least affected	(3) <i>p</i> -value for the difference
Teacher college degree	0.039 (0.019, 0.059)	0.800 (0.780, 0.820)	0.000
Teacher training hours	2.447 (2.399, 2.494)	1.684 (1.636, 1.731)	0.000
Teacher ranking	0.666 (0.635, 0.697)	0.405 (0.374, 0.437)	0.000
Student age	14.18 (14.11, 14.25)	13.73 (13.65, 13.80)	0.000
Teacher experience (years)	16.18 (15.60, 16.76)	13.16 (12.58, 13.74)	0.000
Student female	0.417 (0.385, 0.449)	0.555 (0.523, 0.587)	0.000
Teacher age	37.51 (37.02, 38.00)	35.01 (34.52, 35.50)	0.000
Student math score at baseline	− 0.029 (− 0.088, 0.031)	0.169 (0.110, 0.229)	0.005
Student baseline math anxiety	0.298 (0.236, 0.360)	− 0.219 (− 0.281, − 0.157)	0.000
Class size	52.87 (51.82, 53.93)	64.37 (63.32, 65.43)	0.000

Notes: This table shows the average value of the teacher and student characteristics for the most and least affected groups. The estimates are obtained using neural network to produce the proxy predictor $S(Z)$. 90% confidence intervals are reported in parenthesis. The variables *Student math score at baseline* and *Student baseline math anxiety* are normalized. The values reported correspond to the medians over 100 splits.

training that the teacher received prior to the intervention, which is not found to be a determinant of heterogeneity in the original paper, is higher in the most affected group compared to the least affected group.²⁸ This may reflect the fact that teachers who have had more training in the past may be able to better implement the suggestions from the PD intervention. Table 6 shows that teacher rank, experience, and age are higher in the most affected group compared to the least affected group. This is consistent with the existence of a similar mechanism: teachers who have more experience may be able to better implement the suggestions from the PD intervention. As the PD is mainly theoretical, having had other types of training, or having more experience, may be helpful for an effective implementation of the practices learned during the PD.

We then examine whether any of the *student characteristics* are potential drivers of heterogeneity. In contrast to the findings in Loyalka et al. (2019a), who did not find heterogeneity in terms of student features, we find that students in the most affected group differ in terms of several characteristics compared to students in the least affected group. Among the most correlated with the heterogeneity score (listed in Table 6) are student age and gender: students in the most affected group are on average about half a year older than students in the least affected group,

²⁸ The variable indicating teacher training hours previous to the intervention is a categorical variable, based on the terciles of the continuous variable. As the continuous variable is not included in the replication data set of the original paper, for our analysis we use this categorical variable, which takes values 1 to 3, where 3 is the top tercile in the number of training hours.

and the most affected group includes a larger share of male students. Additionally, students in the most affected group, on average, have a lower baseline math score, and tend to be more anxious about math. Thus, teacher PD could be more beneficial for weaker students, and for students who are more anxious about the subject. Finally, class size appears to be a possible determinant of heterogeneity: students who benefit more from the PD tend to be in smaller classes. This result suggests that in smaller classes it may be easier for teachers to implement some of the practices introduced during the PD training. For instance, Loyalka et al. (2019a) mention having students work together in small groups as one of the techniques that were suggested in the PD; this technique is likely to be easier to implement in smaller classes.

In conclusion, our analysis confirms the presence of heterogeneous effects of the teacher PD intervention, and uncovers a rich set of potential determinants of heterogeneity. With the GATES analysis, we are able to show that the achievement of students belonging to the bottom quintile is negatively affected by the intervention, while the achievement of students in the top quintile is positively affected by the intervention. This confirms what was suggested by Loyalka et al. (2019a): that there is a group of students who benefits from the intervention, and a group who does not. In addition, the GATES analysis shows that the effect is not significantly different from zero for the students belonging to the middle quintiles. With the CLAN analysis, we can obtain a clearer picture of the characteristics of the groups who benefit and who do not from the intervention, compared to the original HTE analysis. In line with Loyalka et al. (2019a), we find that teacher characteristics such as having a college degree or having a major in math are potential determinants of heterogeneity. However, our study uncovers additional differences (that were not identified in the original paper) between the least and the most affected groups, in terms of both teacher and student characteristics, such as teacher's rank, experience, age, and number of training hours, as well as student's gender, age, baseline math score, baseline math anxiety, and class size.

4. CONCLUSION

Our main message is that appropriately combining predictive methods with causal questions adds value to traditional methods and should be more often explored in applied research. We argue that in each revisited study the researcher would have benefited from employing causal ML methods and would have gained additional insights not provided by standard causal inference tools.

We offer the following recommendations for applied researchers about the usefulness of causal ML methods.

- (a) Causal ML methods are useful in settings with many potential confounders relative to the sample size. With our revisited examples, we show the importance of taking into account all potentially relevant confounders at once, both linearly and nonlinearly. We revisit the most complete robustness checks of Djankov et al. (2010a) and Nunn and Treffer (2010a), considering all potential confounders linearly and nonlinearly, which would not be possible with traditional methods. Furthermore, our results from the Monte Carlo study suggest that, as the number of covariates used in the estimation increases relative to the sample size, the gains from using DML over OLS increase.
- (b) Causal ML methods are more suitable than traditional methods to flexibly capture the effect of covariates. As the true functional form is unknown, with flexible estimation we can better capture the effect of confounders. For instance, when revisiting the results of

Djankov et al. (2010a), we show suggestive evidence that there exist relevant nonlinear terms, which were not taken into account in the original analysis, but are captured by the DML estimation. Moreover, we show with our MC simulations that in the presence of nonlinear confounders DML outperforms OLS.

- (c) We further recommend using causal ML methods in settings where the researcher does not have a lot of guidance from theory on which covariates should be included. This is because they implement a systematic model selection, rather than choosing an ad hoc specification, as we discuss when revisiting the results of Djankov et al. (2010a). This argument is also important when performing sensitivity analysis and robustness checks, as highlighted by our results when revisiting Nunn and Treffer (2010a).
- (d) Finally, if the researcher is interested in HTE, causal machine learning methods can ensure that relevant heterogeneity and its determinants are not missed, or falsely discovered due to multiple hypothesis testing issues. For example, our analysis of the HTEs of the papers by Della Vigna and Kaplan (2007a) and Loyalka et al. (2019a) reveals potential determinants of heterogeneity that were not considered in the original analyses, which rely on traditional methods. Moreover, causal ML methods can be used to uncover heterogeneity *ex post*, without being bound to explore HTE only for the specific subgroups indicated in the pre-analysis plan.

These advantages are particularly important in the context of observational studies, where causal ML methods can improve the credibility of causal analysis by making the unconfoundedness assumption more plausible. However, even if the empirical study is a randomized control trial and controlling for confounding factors is not necessarily needed, the use of causal machine learning methods can improve efficiency and provide more precise estimates with lower standard errors and tighter confidence intervals.

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